

Work Stream 4: Mechanically Driven Cavitation

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1 Introduction

A fascinating question posed by [u/Little-Name9809](#) is whether the mathematical construction of the OpenAI singularity suggests a physical mechanism to vaporize a fluid into plasma using *only mechanical force* (with no external heat).

2 Hydrodynamic Cavitation

In the OpenAI proof, the internal wave packets transfer energy into the vortex core at a rate that diverges as $\tau \rightarrow 0$. By evaluating the local cavitation number σ :

$$\sigma(\tau) = \frac{p_\infty - p_v}{\frac{1}{2}\rho|u(\tau)|^2} \quad (1)$$

where p_v is the vapor pressure. As the core velocity $|u(\tau)| \sim \tau^{-1/2} \rightarrow \infty$, the cavitation number rapidly approaches zero.

Before the thermal shock discussed in Work Stream 2 boils the fluid, the extreme drop in local static pressure (due to Bernoulli effects in the vortex core) induces *mechanical cavitation*. This is analogous to sonoluminescence, where violent bubble collapse generates localized plasma. The mathematical singularity provides a theoretical blueprint for maximizing this acoustic concentration of energy.

3 Acknowledgments

Thank you to [u/Little-Name9809](#) for connecting the pure math PDE construction to the physics of sonoluminescence and phase transitions.